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Education: Ph.D. The Johns Hopkins University
M.A. The Cornell University
B.A. The National Cheng Kung University

Current Editorship: Associate Editor
Statistics and Probability letters

Associate Editor
Methodology and Computing in Applied Probability

Research Interests: Large sample estimation, Distribution theory of runs and patterns, Apply probability, System reliability, and Computer intensive statistical methods

Professional Award (Sept. 2001):

ISI Citation Classic Award for the paper: Fu, James C. (1996). Distribution theory of runs and patterns associated with a sequence of multi-state trials. *Statistica Sinica*. **6**, 957-974.

Recent Publications (2000-2005):

Book:

[1] Fu, James, C. and Lou, W.Y. Wendy (2003). *Distributions Theory of Runs and Patterns and its Applications: a Finite Markov Chain Imbedding Approach*. World Scientific.

Journal Publication (1998-2003):

[1] Fu, J.C. and Lou, W.Y.W. (2005). Distribution of the length of the longest common subsequence of two multi-state biological sequences. *Journal of Statistical Planning and Inference*. (to appear)

[2] Fu, J.C. and Lou, W.Y.W. (2005). Waiting time distribution of simple and compound patterns in a sequence of r-th order Markov dependent multi-state trials. *Ann. Inst. Statist. Math.* (to appear)

[3] Wang, L and Fu, J.C. (2005). A practical sampling approach for a Bayesian mixture model with unknown number of parameters. *Statistical paper*. (to appear)

[4] Xue, L., Fu, J. C., Wang, F., and Wang, L. (2005). A mixture model approach to Analyzing major element chemistry data of Changjiang (Yangtze River). *Environmetric*. **16**, 305-318.

[5] Fu, James C. and Chang, Y.M. (2003). On ordered series and later waiting time distributions in a sequence of Markov dependent multi-state trials. *Journal of Applied Probability*. **40**, 623-642.

[6] Fu, James C., Wang, Liqun and Lou, W.Y. Wendy (2003). On exact and large deviation approximation for the distribution of the longest run in a sequence of two-state Markov dependent trials. *Journal of Applied Probability*. **40**, 346-360.

[7] Fu, James C., Shmueli, Galit and Chang, Y.M. (2003). A unified Markov chain approach for computing the run length distribution in control charts with simple or compound rules. *Statistics and Probability Letters*. **65**, 457-466.

- [8] Fu, James C. and Wang, Liqun (2002). A random discretization based Monte Carlo sampling method and its applications. *Methodology and computing in Applied Probability*. **4**, 5-25.
- [9] Fu, James C. and Chang Y.M. (2002). On probability generating functions for waiting time distributions of Compound patterns in a sequence of multistate trials. *Journal of Applied Probability*. **39**, 70-80.
- [10] Fu, James C., Lou, W.Y. Wendy, Bai, Zhi-Dong and Li, Gang (2002). The exact and limiting distribution for the number of successes in success runs within a sequence of Markov dependent two-state trials. *Annals of the Institute of Statistical Mathematics*. **54**, 719-730.
- [11] Fu, James C., Spiring, Fred A. and Xie, Hansheng (2002). On the average run lengths of quality control schemes using a Markov chain approach. *Statistics and Probability Letters*. **56**, 369-380.
- [12] Fu, James C. (2001). Distributions of the scan statistic for a sequence of bivariate trials. *Journal of Applied probability*. **38**, 908-916.
- [13] Fu, James C.(2001). On sensitivity of exponential rate of convergence for the maximum likelihood estimator. *Probability and Statistical Models with Applications*. Chapman and Hall, 429-446.
- [14] Fu, James C. and Lou, W.Y. Wendy (2000). Joint distributions of rises and falls. *Annals of the Institute of Statistical Mathematics*. **52**, 415-425.
- [15] Johnson, B.C. and Fu, James C. (2000). The distribution of L-sequence in random permutations: A Markov Chain approach. *Statistics and Probability Letters*. **49**, 337-344.
- [16] Fu, J.C. and Lou, W.Y.W. (2000). On the exact distribution of SECON and its application. *Statistica Sinica*. **10**. 999-1010.

Invited Lecture Tour (July 1 to 15, 2003)

Institute of Mathematics and Systems Sciences, Chinese Academy of Sciences, Beijing, China

Topic: Distribution Theory of Runs and Patterns: A Finite Markov Chain Imbedding Approach